

Package ‘ppmSDR’

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Type Package

Title Penalized Principal Machine for Sufficient Dimension Reduction

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Description A unified, computation-friendly framework for penalized principal machines (P2M), a class of sparse sufficient dimension reduction (SDR) estimators for regression and binary classification. Principal machines (PM) estimate the central subspace by solving a family of convex-loss problems over several cutoffs; their penalized counterparts (P2M) add a row-group sparsity penalty so that dimension reduction and variable selection are performed simultaneously. All estimators are fitted by a single group coordinate descent (GCD) algorithm that accommodates least squares, logistic, asymmetric least squares, L2-hinge, hinge (support vector machine, SVM) and quantile losses, together with the least absolute shrinkage and selection operator (LASSO), the smoothly clipped absolute deviation (SCAD) penalty and the minimax concave penalty (MCP). Methods are described in Li, Artemiou and Li (2011) <doi:10.1214/11-AOS932>, Shin and Artemiou (2017) <doi:10.1016/j.csda.2016.12.003>, Artemiou, Dong and Shin (2021) <doi:10.1016/j.patcog.2020.107768> and Breheny and Huang (2015) <doi:10.1007/s11222-013-9424-2>.

Depends R (>= 4.1.0)

License GPL-3

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Imports stats, Matrix, grpreg, energy

Suggests MASS, knitr, rmarkdown, testthat (>= 3.0.0)

URL <https://github.com/c16267/ppmSDR>

BugReports <https://github.com/c16267/ppmSDR/issues>

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boston	<i>Boston Housing Data</i>
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Description

The Boston housing data of Harrison and Rubinfeld (1978), as distributed in **MASS**. Following the real-data analysis in the package’s accompanying article, the binary Charles-River dummy variable (chas) is removed, leaving twelve continuous predictors and the response medv. Values are on their natural scale; standardize them before fitting (see Examples).

Usage

boston

Format

- A data frame with 506 rows and 13 variables:
- crim** per-capita crime rate by town.
 - zn** proportion of residential land zoned for lots over 25,000 sq ft.
 - indus** proportion of non-retail business acres per town.
 - nox** nitrogen-oxides concentration (parts per 10 million).
 - rm** average number of rooms per dwelling.
 - age** proportion of owner-occupied units built prior to 1940.
 - dis** weighted mean of distances to five Boston employment centres.
 - rad** index of accessibility to radial highways.
 - tax** full-value property-tax rate per \$10,000.
 - ptratio** pupil-teacher ratio by town.
 - b** $1000(Bk - 0.63)^2$, where Bk is the proportion of the Black population by town. See the note below.
 - lstat** lower-status percentage of the population.
 - medv** median value of owner-occupied homes in \$1000s (the response).

Note

The variable **b** embeds a racist premise from the original 1978 study and is retained only to reproduce the published analysis. It should not be used uncritically; see the discussion in the documentation of **MASS::Boston**.

Source

Harrison, D. and Rubinfeld, D. L. (1978) Hedonic prices and the demand for clean air. *Journal of Environmental Economics and Management*, 5, 81–102. Distributed in the **MASS** package.

Examples

```
data(boston)
x <- scale(as.matrix(boston[, setdiff(names(boston), "medv")]))
y <- as.numeric(scale(boston$medv))
fit <- ppm(x, y, loss = "lssvm", penalty = "grSCAD", lambda = 8e-3)
summary(fit, d = 2)
```

ppm

Penalized Principal Machine for Sufficient Dimension Reduction

Description

Fits a penalized principal machine (P2M), a sparse sufficient dimension reduction (SDR) estimator, through a single group coordinate descent (GCD) engine. A principal machine (PM) estimates the basis of the central subspace by solving a family of convex-loss problems over several cutoffs (slices); the penalized version adds a row-group sparsity penalty so that dimension reduction and variable selection are performed simultaneously.

Usage

```
ppm(
  x,
  y,
  loss = "lssvm",
  H = 10,
  C = 1,
  lambda = 0.01,
  gamma = 3.7,
  penalty = c("grSCAD", "grLasso", "grMCP"),
  max.iter = 100,
  ...
)
```

Arguments

- | | |
|------|---|
| x | A numeric matrix or data frame of predictors, of dimension n (observations) by p (variables). |
| y | A response vector of length n. For continuous-type losses a numeric vector is expected; for weighted losses (those whose name starts with "w") a two-class response is expected and is recoded internally to $\{-1, +1\}$. |
| loss | Character string selecting the loss function. One of "lssvm", "wlssvm", "svm", "wsvm", "l2svm", "wl2svm", "logit", "wlogit", "asls", "qr". Default "lssvm". |
| H | Number of cutoffs (slices); a single integer ≥ 2 . Default 10. |
| C | Positive cost parameter that balances the loss against the covariance term. Default 1. |

<code>lambda</code>	Positive regularization parameter controlling sparsity. Default 0.1. In practice <code>lambda</code> should be selected by cross-validation.
<code>gamma</code>	Concavity parameter of the SCAD/MCP penalty; must exceed 2. Default 3.7.
<code>penalty</code>	Penalty type: "grSCAD" (default), "grLasso" or "grMCP".
<code>max.iter</code>	Maximum number of GCD iterations. Default 100.
<code>...</code>	Additional arguments passed to the underlying solver. The most useful is <code>ridge</code> , a small non-negative ridge constant added for numerical stability, which is accepted by the iterative solvers (<code>logit</code> , <code>wlogit</code> , <code>svm</code> , <code>wsvm</code> , <code>qr</code> , <code>lssvm</code> , <code>wlssvm</code> , <code>wl2svm</code>).

Details

`ppm()` is a single front-end that dispatches to the loss-specific solver selected by `loss`. Two families are supported, following the taxonomy of Shin and Shin (2024):

- **Response-based PM (RPM)** for a continuous response, where the loss is fixed and the pseudo-response varies across slices: "lssvm" (least squares, P2LSM), "l2svm" (L2-hinge, P2L2M), "svm" (hinge, P2SVM), "logit" (logistic, P2LR), "asls" (asymmetric least squares, P2AR) and "qr" (quantile, P2QR).
- **Loss-based PM (LPM)** for a binary response coded internally as $\{-1, +1\}$, where the loss varies across slices: "wlssvm" (P2WLSM), "wl2svm" (P2WL2M), "wsvm" (P2WSVM) and "wlogit" (P2WLR).

Acronyms used above: SDR (sufficient dimension reduction), PM (principal machine), P2M (penalized principal machine), GCD (group coordinate descent), SVM (support vector machine). The `penalty` is one of the group LASSO (least absolute shrinkage and selection operator), the group SCAD (smoothly clipped absolute deviation) or the group MCP (minimax concave penalty), passed as "grLasso", "grSCAD" or "grMCP".

The basis of the central subspace is estimated by the leading eigenvectors of the working matrix $M = \sum_{k=1}^H \beta_k \beta_k^\top$, where β_k is the slope estimated at the k -th cutoff.

Value

An object of S3 class "ppm", a list containing:

M the estimated working matrix (a p by p symmetric matrix).

values, evecs the eigenvalues and eigenvectors of **M**; the leading d eigenvectors estimate the basis of the central subspace.

x, y the (validated) input data.

loss, penalty, lambda, gamma, C, H, max.iter the fitting settings.

ytype "continuous" or "binary".

n, p the sample size and the number of predictors.

call the matched call.

References

- Li, B., Artemiou, A. and Li, L. (2011) Principal support vector machines for linear and nonlinear sufficient dimension reduction. *The Annals of Statistics*, 39(6), 3182–3210. doi:10.1214/11AOS932
- Shin, S. J. and Artemiou, A. (2017) Penalized principal logistic regression for sparse sufficient dimension reduction. *Computational Statistics & Data Analysis*, 111, 48–58. doi:10.1016/j.csda.2016.12.003

Breheny, P. and Huang, J. (2015) Group descent algorithms for nonconvex penalized linear and logistic regression models with grouped predictors. *Statistics and Computing*, 25, 173–187. doi:10.1007/s1122201394242

See Also

`print.ppm()`, `summary.ppm()`

Examples

```
set.seed(1)
n <- 1000; p <- 10
B <- matrix(0, p, 2); B[1, 1] <- B[2, 2] <- 1
x <- matrix(rnorm(n * p), n, p)
y <- (x %*% B[, 1]) / (0.5 + (x %*% B[, 2] + 1)^2) + 0.2 * rnorm(n)

## penalized principal least-squares SVM (P2LSM) with the group SCAD penalty
fit <- ppm(x, y, loss = "lssvm", penalty = "grSCAD", lambda = 0.01)
round(fit$sevectors[, 1:2], 3)
print(fit)
summary(fit)

## binary response: penalized principal asymmetric least squares (P2AR)
yb <- sign(y)
fitw <- ppm(x, yb, loss = "asls", penalty = "grSCAD", lambda = 0.04)
round(fitw$sevectors[, 1:2], 3)
print(fitw)
summary(fitw)

data(boston)
xb <- scale(as.matrix(boston[, setdiff(names(boston), "medv")]))
yb <- as.numeric(scale(boston$medv))
fit_b <- ppm(xb, yb, loss = "lssvm", penalty = "grSCAD", lambda = 8e-3)
summary(fit_b, d = 2)
```

ppm_tune

Cross-Validation for the Penalized Principal Machine Tuning Parameter

Description

Selects the sparsity parameter λ of a penalized principal machine by K -fold cross-validation. For each candidate λ , the model is fitted on the training folds and the held-out distance correlation (dCor; Szekely, Rizzo and Bakirov, 2007) between the response and the predictors projected onto the estimated d -dimensional central subspace is recorded. The λ maximizing the average held-out dCor is returned, following the criterion of Shin, Wu, Zhang and Liu (2017).

Usage

```
ppm_tune(
  x,
  y,
  loss = "lssvm",
  d = 2,
  H = 10,
  C = 1,
  gamma = 3.7,
  penalty = c("grSCAD", "grLasso", "grMCP"),
  max.iter = 100,
  n.fold = 5,
  lambda = NULL,
  nlambdas = 20,
  lambda.max = 0.1,
  lambda.min.ratio = 0.001,
  verbose = FALSE,
  ...
)
```

Arguments

<code>x</code>	A numeric matrix or data frame of predictors, of dimension n (observations) by p (variables).
<code>y</code>	A response vector of length n . For continuous-type losses a numeric vector is expected; for weighted losses (those whose name starts with "w") a two-class response is expected and is recoded internally to $\{-1, +1\}$.
<code>loss</code>	Character string selecting the loss function. One of "lssvm", "wlssvm", "svm", "wsvm", "l2svm", "wl2svm", "logit", "wlogit", "asls", "qr". Default "lssvm".
<code>d</code>	Working structural dimension used to form the held-out sufficient predictors. Default 2.
<code>H</code>	Number of cutoffs (slices); a single integer ≥ 2 . Default 10.
<code>C</code>	Positive cost parameter that balances the loss against the covariance term. Default 1.
<code>gamma</code>	Concavity parameter of the SCAD/MCP penalty; must exceed 2. Default 3.7.
<code>penalty</code>	Penalty type: "grSCAD" (default), "grLasso" or "grMCP".
<code>max.iter</code>	Maximum number of GCD iterations. Default 100.
<code>n.fold</code>	Number of cross-validation folds. Default 5. For reproducible folds, call <code>set.seed()</code> before <code>ppm_tune()</code> .
<code>lambda</code>	Optional numeric vector of candidate values. If NULL (default), a log-spaced grid of length <code>nlambdas</code> is built from <code>lambda.max</code> down to <code>lambda.max * lambda.min.ratio</code> .
<code>nlambdas</code>	Length of the automatically generated grid. Default 20.
<code>lambda.max</code>	Largest candidate value in the generated grid. Default 1.
<code>lambda.min.ratio</code>	Ratio of the smallest to the largest candidate in the generated grid. Default $1e-3$.
<code>verbose</code>	Logical; if TRUE, progress is reported via <code>message()</code> . Default FALSE.
<code>...</code>	Additional arguments passed to the underlying solver. The most useful is <code>ridge</code> , a small non-negative ridge constant added for numerical stability, which is accepted by the iterative solvers (logit, wlogit, svm, wsvm, qr, lssvm, wlssvm, wl2svm).

Value

An object of S3 class "ppm_tune", a list with elements

opt.lambda the selected value of lambda.

lambda the candidate grid (decreasing).

dcor the average held-out distance correlation at each candidate.

fit a `ppm()` object refitted on all data at `opt.lambda`.

d, loss, penalty, n.fold, call the settings and the matched call.

References

Shin, S. J., Wu, Y., Zhang, H. H. and Liu, Y. (2017) Principal weighted support vector machines for sufficient dimension reduction in binary classification. *Biometrika*, 104(1), 67–81. doi:[10.1093/biomet/asw057](https://doi.org/10.1093/biomet/asw057)

Szekely, G. J., Rizzo, M. L. and Bakirov, N. K. (2007) Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, 35(6), 2769–2794. doi:[10.1214/0090536070000000505](https://doi.org/10.1214/0090536070000000505)

See Also

`ppm()`

Examples

```
set.seed(1)
n <- 1000; p <- 10
x <- matrix(rnorm(n * p), n, p)
y <- x[, 1] / (0.5 + (x[, 2] + 1)^2) + 0.2 * rnorm(n)
cv <- ppm_tune(x, y, loss = "lssvm", d = 2, n.fold = 5)
cv$opt.lambda
summary(cv$fit, d = 2)
```

print.ppm

Print a fitted penalized principal machine

Description

Print a fitted penalized principal machine

Usage

```
## S3 method for class 'ppm'
print(x, ...)
```

Arguments

x An object of class "ppm" returned by `ppm()`.

... Currently ignored.

Value

The input object x, invisibly.

See Also

[ppm\(\)](#), [summary.ppm\(\)](#)

Examples

```
set.seed(1)
x <- matrix(rnorm(200 * 6), 200, 6)
y <- x[, 1] / (0.5 + (x[, 2] + 1)^2) + 0.2 * rnorm(200)
fit <- ppm(x, y, loss = "lssvm", lambda = 0.01)
print(fit)
```

```
print.ppm_tune
```

```
Print a penalized principal machine cross-validation result
```

Description

Print a penalized principal machine cross-validation result

Usage

```
## S3 method for class 'ppm_tune'
print(x, ...)
```

Arguments

x An object of class "ppm_tune" returned by [ppm_tune\(\)](#).
 ... Currently ignored.

Value

The input object x, invisibly.

See Also

[ppm_tune\(\)](#)

summary.ppm

*Summarize a fitted penalized principal machine***Description**

Reports the estimated basis of the central subspace for a given working dimension d , together with the predictors selected by the row-group penalty (those whose loadings are non-zero across the leading d directions).

Usage

```
## S3 method for class 'ppm'
summary(object, d = 2, tol = 1e-06, ...)
```

Arguments

object	An object of class "ppm" returned by <code>ppm()</code> .
d	Working structural dimension; the number of leading eigenvectors to report. Default 2.
tol	Tolerance below which a row L2-norm is treated as zero when determining the selected variables. Default 1e-6.
...	Currently ignored.

Value

Invisibly, a list with elements `d`, `basis` (the p by d estimated basis), `selected` (indices of the selected predictors) and `n.selected`.

See Also

`ppm()`, `print.ppm()`

Examples

```
set.seed(1)
x <- matrix(rnorm(200 * 6), 200, 6)
y <- x[, 1] / (0.5 + (x[, 2] + 1)^2) + 0.2 * rnorm(200)
fit <- ppm(x, y, loss = "lssvm", lambda = 0.01)
summary(fit, d = 2)
```

wdbc

*Wisconsin Diagnostic Breast Cancer (WDBC) Data***Description**

Diagnostic measurements for 569 breast masses from the Wisconsin Diagnostic Breast Cancer study (Street, Wolberg and Mangasarian; 1993). Each record has thirty real-valued features computed from a digitized image of a fine-needle aspirate, namely the mean, standard error (`_se`) and worst (largest) value of ten cell-nucleus characteristics, together with a benign/malignant diagnosis. Features are on their natural scale; standardize them before fitting (see Examples).

Usage

wdbc

Format

A data frame with 569 rows and 31 variables: a factor diagnosis with levels "B" (benign, 357 cases) and "M" (malignant, 212 cases), followed by thirty numeric features. The features are the `_mean`, `_se` and `_worst` summaries of: radius, texture, perimeter, area, smoothness, compactness, concavity, concave_points, symmetry and fractal_dimension (for example `radius_mean`, `radius_se`, `radius_worst`).

Details

For the weighted-loss principal machines ("`wsvm`", "`wlssvm`", "`wlogit`", "`wl2svm`") the response is coded as malignant = +1 and benign = -1, as shown in the Examples.

Source

Street, W. N., Wolberg, W. H. and Mangasarian, O. L. (1993) Nuclear feature extraction for breast tumor diagnosis. *Biomedical Image Processing and Biomedical Visualization*, 1905, 861–870. UCI Machine Learning Repository, Breast Cancer Wisconsin (Diagnostic) Data Set.

Examples

```
data(wdbc)
x <- scale(as.matrix(wdbc[, -1]))
y <- ifelse(wdbc$diagnosis == "M", 1, -1)
fit <- ppm(x, y, loss = "w12svm", penalty = "grSCAD", lambda = 0.3)
summary(fit, d = 2)
B <- fit$evectors[, 1:2]
scores <- x %*% B
plot(scores[, 1], scores[, 2], col = ifelse(y == 1, "red", "blue"),
     pch = ifelse(y == 1, 17, 1),
     xlab = "1st SDR direction", ylab = "2nd SDR direction")
legend("topright", legend = c("malignant (+1)", "benign (-1)"),
     col = c("red", "blue"), pch = c(17, 1))
```

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